

2. Linear Regression

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2.1 Model & Normal Equation

- **Model:** $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_d X_d + \varepsilon$ for an $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ independent of $(X_1, \dots, X_d)$
- Let $(x_{i1}, \dots, x_{id}, y_i, \varepsilon_i), i = 1, \dots, m$, be iid realizations of $(X_1, \dots, X_d, Y, \varepsilon)$
- **Matrix notation:** $y = A\beta + \varepsilon$, where

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, A = \underbrace{\begin{pmatrix} 1 & x_{11} & \dots & x_{1d} \\ 1 & x_{21} & \dots & x_{2d} \\ \vdots & \vdots & \dots & \vdots \\ 1 & x_{m1} & \dots & x_{md} \end{pmatrix}}_{\text{design matrix}}, \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_d \end{pmatrix}, \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_m \end{pmatrix}$$

typically, $m \geq d + 1$ (more observations than parameters)

- **Empirical loss minimization with the square loss function = ordinary least squares:**

$$\min_{b \in \mathbb{R}^{d+1}} \|A b - y\|_2^2 = \min_{b \in \mathbb{R}^{d+1}} \sum_{i=1}^m \left(\sum_{j=0}^d x_{ij} b_j - y_i \right)^2 \quad \text{where } x_{i0} = 1, i = 1, \dots, m$$

- Since $\|A b - y\|_2^2$ is **convex** in $b \in \mathbb{R}^{d+1}$, b is a minimizer $\Leftrightarrow \nabla_b \|A b - y\|_2^2 = 0$

$$\Leftrightarrow 0 = \frac{\partial}{\partial b_k} \sum_{i=1}^m \left(\sum_{j=0}^d x_{ij} b_j - y_i \right)^2 = \sum_{i=1}^m 2 \left(\sum_{j=0}^d x_{ij} b_j - y_i \right) x_{ik} \quad \text{for all } k = 0, 1, \dots, d$$

$$\Leftrightarrow A^T A b = A^T y \quad (\text{normal equation})$$

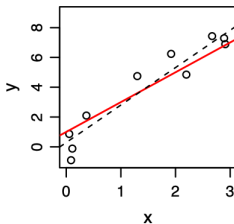
Solving the normal equation $A^T A b = A^T y$

- If the columns of A are linearly independent, $A^T A$ is regular, and the unique solution of $A^T A b = A^T y$ is

$$\hat{\beta} = (A^T A)^{-1} A^T y$$

depends on the data

$$x_{ij}, y_i, \quad i = 1, \dots, m, \\ j = 0, \dots, d$$



red = true regression line

dashed = estimated regression line

- Conditioned on $X_{ij} = x_{ij}$, $i = 1, \dots, m$, $j = 1, \dots, d$,

$$\hat{\beta} = (A^T A)^{-1} A^T (A\beta + \epsilon) = \beta + (A^T A)^{-1} A^T \epsilon \sim \mathcal{N}_{d+1} \left(\beta, \sigma^2 (A^T A)^{-1} \right)$$

since $\text{Cov}((A^T A)^{-1} A^T \epsilon) = \mathbb{E}(A^T A)^{-1} A^T \epsilon \epsilon^T A (A^T A)^{-1} = \sigma^2 (A^T A)^{-1} A^T A (A^T A)^{-1} = \sigma^2 (A^T A)^{-1}$

in particular, $\hat{\beta}$ is unbiased

- $A^T A$ is always a symmetric positive semidefinite $l \times l$ -matrix for $l = d + 1$
 $\Rightarrow A^T A = V \Lambda V^T$ for an orthogonal $V \in \mathbb{R}^{l \times l}$ and a diagonal $\Lambda \in \mathbb{R}^{l \times l}$ with diagonal entries
 $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_l \geq 0$

- **Pseudoinverse:** $(A^T A)^\dagger := V \Lambda^\dagger V^T$, where $\Lambda^\dagger$ is diagonal with diagonal entries
 $1_{\{\lambda_j > 0\}} \lambda_j^{-1}, j = 1, \dots, l$

Lemma If the columns of A are linearly dependent, the normal equation $A^T A b = A^T y$ has several solutions; $\hat{\beta} = (A^T A)^\dagger A^T y \in \mathbb{R}^l$ is the solution with the smallest $\|\cdot\|_2$ -norm

$(A^\dagger := (A^T A)^\dagger A^T \in \mathbb{R}^{l \times m}$ is called *pseudoinverse of A*)

Proof

1. Denote the rank of A by $r \leq \min \{m, l\}$ (it is also the rank of A^T).

If $A^T A v = 0$, then $\|Av\|_2^2 = v^T A^T A v = 0 \Rightarrow Av = 0$, that is, $\text{Ker}(A^T A) = \text{Ker}(A)$
 $\Rightarrow \dim(\text{Im}(A^T A)) = l - \dim(\text{Ker}(A^T A)) = l - \dim(\text{Ker}(A)) = r = \dim(\text{Im}(A^T))$.

Since $\text{Im}(A^T A) \subseteq \text{Im}(A^T)$, this shows that $\text{Im}(A^T A) = \text{Im}(A^T)$.

Moreover, one has $A^T A = V \Lambda V^T$ and $\lambda_1 \geq \dots \geq \lambda_r > 0$, $\lambda_{r+1} = \dots = \lambda_l = 0$.

Therefore, $\text{Im}(A^T) = \text{Im}(A^T A) = \text{span}(v_1, \dots, v_r)$, where $v_1, \dots, v_r$ are the first r columns of V .

In particular, $A^T y = \sum_{i=1}^r a_i v_i$ for some coefficients $a_1, \dots, a_r \in \mathbb{R}$.

If $P_r : \mathbb{R}^l \rightarrow \mathbb{R}^l$ is the orthogonal projection on the first r coordinates,

$Q_r = V P_r V^T$ is the orthogonal projection on $\text{span}(v_1, \dots, v_r)$

$\Rightarrow A^T A \hat{\beta} = A^T A (A^T A)^\dagger A^T y = V \Lambda V^T V \Lambda^\dagger V^T A^T y = V P_r V^T A^T y = Q_r A^T y = A^T y$,
that is, $\hat{\beta}$ solves the normal equation.

2. $\hat{\beta} = (A^T A)^\dagger A^T y = V \Lambda^\dagger V^T A^T y \in \text{span}(v_1, \dots, v_r)$

If $A^T A(\hat{\beta} + b) = A^T y$, then $b \in \text{Ker}(A^T A) = \text{span}(v_{r+1}, \dots, v_l)$

$$\text{and } \|\hat{\beta} + b\|_2^2 = \|\hat{\beta}\|_2^2 + \|b\|_2^2.$$

So, $\hat{\beta}$ is the solution of the normal equation with the smallest $\|\cdot\|_2$ -norm

□

2.2 Singular Value Decomposition

Theorem

Let A be an $m \times l$ -matrix of rank $r \leq \min\{m, l\}$. Then ...

- ① there exists an orthonormal basis $v_1, \dots, v_l$ of $\mathbb{R}^l$ consisting of eigenvectors of $A^T A$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$ and $\lambda_{r+1} = \dots = \lambda_l = 0$
- ② for all $i = 1, \dots, l$, Av_i is an eigenvector of AA^T with eigenvalue λ_i
- ③ $u_i = \lambda_i^{-1/2} A v_i$, $i = 1, \dots, r$, are orthonormal eigenvectors of AA^T with eigenvalues λ_i , $i = 1, \dots, r$
- ④ $A = \sum_{i=1}^r \sigma_i u_i v_i^T$ for $\sigma_i = \sqrt{\lambda_i}$, $i = 1, \dots, r$ (SVD of A)
- ⑤ the *pseudoinverse* of A is given by $A^\dagger = \sum_{i=1}^r \sigma_i^{-1} v_i u_i^T$
- ⑥ In particular, for any $y \in \mathbb{R}^m$, $\hat{\beta} = A^\dagger y \in \mathbb{R}^n$ is the minimizer of $b \mapsto \|A b - y\|_2$ with the minimal $\|\cdot\|_2$ -norm

$\sigma_1 \geq \dots \geq \sigma_r > 0$ and $\sigma_{r+1} = \dots = \sigma_l = 0$ are the *singular values* of A
(they are unique)

Proof

1. $A^T A$ is symmetric and positive semidefinite. Moreover, we know that $\text{rank}(A^T A) = \text{rank}(A) = r$. Therefore, there exists an orthonormal basis $v_1, \dots, v_l$ of $\mathbb{R}^l$ consisting of eigenvectors of $A^T A$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$ and $\lambda_{r+1} = \dots = \lambda_l = 0$

2. $AA^T A v_i = A \lambda_i v_i = \lambda_i A v_i$

3. $u_i^T u_j = \lambda_i^{-1/2} (A v_i)^T \lambda_j^{-1/2} A v_j = \lambda_i^{-1/2} \lambda_j^{-1/2} v_i^T A^T A v_j = \lambda_i^{-1/2} \lambda_j^{-1/2} v_i^T (\lambda_j v_j) = \delta_{ij}$

4. $A v_i = \sigma_i u_i$ for $i = 1, \dots, r$, and $\|A v_i\|_2^2 = v_i^T A^T A v_i = 0$ for $i = r+1, \dots, l$

$$\Rightarrow A = \sum_{i=1}^r \sigma_i u_i v_i^T$$

5. $A^\dagger := (A^T A)^\dagger A^T = \left(\sum_{i=1}^r \lambda_i^{-1} v_i v_i^T \right) \left(\sum_{i=1}^r \sigma_i v_i u_i^T \right) = \sum_{i=1}^r \sigma_i^{-1} v_i u_i^T$

6. follows from the last proof



Corollary Let A be an $m \times l$ -matrix of rank $r \leq \min\{m, l\}$. Then it has a decomposition of the form $A = U\Sigma V^T$ (also called **SVD**) and $A^\dagger = V\Sigma^\dagger U^T$, where ...

- U is an orthogonal $m \times m$ -matrix
- V is an orthogonal $l \times l$ -matrix
- Σ is a diagonal $m \times l$ -matrix and $\Sigma^\dagger$ a diagonal $l \times m$ -matrix of the forms

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \dots & \dots & \dots & 0 \\ 0 & \sigma_2 & \dots & \dots & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \dots & \dots & \sigma_r & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & \dots & 0 \end{pmatrix} \quad \text{and} \quad \Sigma^\dagger = \begin{pmatrix} \sigma_1^{-1} & 0 & \dots & \dots & \dots & 0 \\ 0 & \sigma_2^{-1} & \dots & \dots & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \vdots & \dots & \dots & \sigma_r^{-1} & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & \dots & 0 \end{pmatrix}$$

for $\sigma_i = \sqrt{\lambda_i}$, where $\lambda_1 \geq \dots \geq \lambda_r$ are the positive eigenvalues of $A^T A$.

Proof Let V be the $l \times l$ -matrix with column vectors $v_1, \dots, v_l$,
choose $u_{r+1}, \dots, u_m$ such that $u_1, \dots, u_m$ is an orthonormal basis of $\mathbb{R}^m$,
let U be the $m \times m$ -matrix with column vectors $u_1, \dots, u_m$

□

2.3 Regularization by Truncating the Pseudoinverse

- If the design matrix A has small positive singular values,
 - $\hat{\beta} = A^\dagger y = A^\dagger (A\beta + \varepsilon)$ is sensitive to the noise $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ and
 - the numerical computation of $A^\dagger = \sum_{i=1}^r \sigma_i^{-1} v_i u_i^T$ becomes unstable

- Consider a threshold $c > 0$

- **truncated SVD** $A_c = \sum_{i=1}^r \sigma_i 1_{\{\sigma_i > c\}} u_i v_i^T = \sum_{i=1}^k \sigma_i u_i v_i^T$, where $k = \#\{i : \sigma_i > c\}$
(dimension reduction)

- **truncated pseudoinverse** $A_c^\dagger = \sum_{i=1}^k \sigma_i^{-1} v_i u_i^T$
- Conditioned on $X_{ij} = x_{ij}$, $i = 1, \dots, m$, $j = 1, \dots, d$,

$$\hat{\beta}_c = A_c^\dagger y = A_c^\dagger A\beta + A_c^\dagger \varepsilon = V \Sigma_c^\dagger \Sigma V^T \beta + A_c^\dagger \varepsilon = Q_k \beta + \sum_{i=1}^k \sigma_i^{-1} v_i u_i^T \varepsilon \sim \mathcal{N}_l(Q_k \beta, \sigma^2 V \Lambda_c^\dagger V^T)$$

- The threshold c is a **hyperparameter**, which balances **bias and variance**
- For increasing c , ...
 - the bias $Q_k \beta - \beta$ of $\hat{\beta}_c$ is increasing
 - the Frobenius norm of the covariance matrix $\sigma^2 V \Lambda_c^\dagger V^T$ is decreasing

2.4 Ridge Regression

- Introduce a penalty term to force the estimate $\hat{\beta}$ to have small Euclidean norm:

$$\min_{b \in \mathbb{R}^{d+1}} \left(\|Ab - y\|_2^2 + \lambda \|b\|_2^2 \right) \quad \text{for a hyperparameter } \lambda > 0$$

- b is a minimum $\Leftrightarrow \nabla_b \left(\|Ab - y\|_2^2 + \lambda \|b\|_2^2 \right) = 0 \Leftrightarrow A^T Ab - A^T y + \lambda b = 0$

$$\Leftrightarrow (A^T A + \lambda I) b = A^T y$$

- $\hat{\beta}_\lambda = (A^T A + \lambda I)^{-1} A^T y = \sum_{i=1}^l \frac{1}{\lambda_i + \lambda} v_i v_i^T \sum_{i=1}^r \sigma_i v_i u_i^T y = \sum_{i=1}^r \frac{\sigma_i}{\sigma_i^2 + \lambda} v_i u_i^T y$

- Conditioned on $X_{ij} = x_{ij}$, $i = 1, \dots, m$, $j = 1, \dots, d$:

$$\begin{aligned} \hat{\beta}_\lambda &= (A^T A + \lambda I)^{-1} A^T (A\beta + \epsilon) = \sum_{i=1}^l \frac{1}{\lambda_i + \lambda} v_i v_i^T \left(\sum_{i=1}^r \lambda_i v_i v_i^T \beta + \sum_{i=1}^r \sigma_i v_i u_i^T \epsilon \right) \\ &= \sum_{i=1}^r \frac{\lambda_i}{\lambda_i + \lambda} v_i v_i^T \beta + \sum_{i=1}^r \frac{\sigma_i}{\lambda_i + \lambda} v_i u_i^T \epsilon \sim \mathcal{N}_l \left(\sum_{i=1}^r \frac{\lambda_i}{\lambda_i + \lambda} v_i v_i^T \beta, \sigma^2 \sum_{i=1}^r \frac{\lambda_i}{(\lambda_i + \lambda)^2} v_i v_i^T \right) \end{aligned}$$

- For increasing λ , ...

- the bias $\sum_{i=1}^r \frac{\lambda_i}{\lambda_i + \lambda} v_i v_i^T \beta - \beta$ of $\hat{\beta}_\lambda$ is increasing
- the Frobenius norm of the covariance matrix $\sigma^2 \sum_{i=1}^r \frac{\lambda_i}{(\lambda_i + \lambda)^2} v_i v_i^T$ is decreasing

Usually ...

one does not penalize the intercept in ridge regression since it represents the baseline level of the response.

That is, one solves the regularized problem

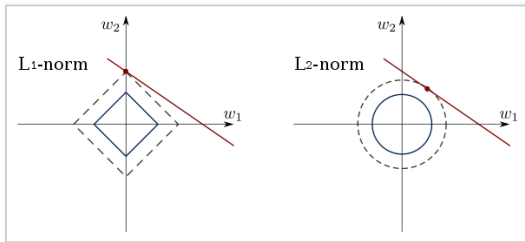
$$\min_{b \in \mathbb{R}^{b+1}} \left\{ \|Ab - y\|_2^2 + \lambda \sum_{j=1}^d b_j^2 \right\} \quad \text{for a hyperparameter } \lambda > 0$$

2.5 LASSO (Least absolute shrinkage and selection operator)

- Introduce a penalty term to force the estimate $\hat{\beta}$ to have small l^1 -norm:

$$\min_{b \in \mathbb{R}^{d+1}} \left(\|Ab - y\|_2^2 + \lambda \sum_{j=1}^d |b_j| \right) \quad \text{for a hyperparameter } \lambda > 0$$

- This is a non-smooth convex optimization problem
- The l^1 -penalty tends to shrink coefficients to 0 and therefore, leads to sparse solutions $\hat{\beta}_\lambda$



2.6 Variable Standardization

- Note that if $b = (b_0, b_1, \dots, b_d)^T$ solves $A^T A b = A^T y$, the first line of the vector equation gives

$$mb_0 + \sum_{j=1}^d \sum_{i=1}^m x_{ij} b_j = \sum_{i=1}^m y_i \quad \Leftrightarrow \quad b_0 = \bar{y} - \sum_{j=1}^d \bar{x}_j b_j$$

$$\text{for } \bar{y} = \frac{1}{m} \sum_{i=1}^m y_i \quad \text{and} \quad \bar{x}_j = \frac{1}{m} \sum_{i=1}^m x_{ij}, \quad j = 1, \dots, d$$

- In particular, if $\bar{x}_1 = \dots = \bar{x}_d = \bar{y} = 0$, then $b_0 = 0$, so the first columns of 1's in A is not needed

- Standardization:** $\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \quad \tilde{y}_i = \frac{y_i - \bar{y}}{s_y}, \quad i = 1, \dots, m, \quad j = 1, \dots, d$

$$\text{for } s_j^2 = \frac{1}{m} \sum_{i=1}^m (x_{ij} - \bar{x}_j)^2, \quad j = 1, \dots, d, \quad \text{and} \quad s_y^2 = \frac{1}{m} \sum_{i=1}^m (y_i - \bar{y})^2$$

- Standardized linear regression:** $\min_{b \in \mathbb{R}^d} \|\tilde{A}b - \tilde{y}\|_2^2$ for $\tilde{y} = \begin{pmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_m \end{pmatrix}$ and $\tilde{A} = \begin{pmatrix} \tilde{x}_{11} & \dots & \tilde{x}_{1d} \\ \tilde{x}_{21} & \dots & \tilde{x}_{2d} \\ \vdots & \dots & \vdots \\ \tilde{x}_{m1} & \dots & \tilde{x}_{md} \end{pmatrix}$

- For a minimizer $\hat{\beta}$ of $\min_{b \in \mathbb{R}^d} \|\tilde{A}b - \tilde{y}\|_2^2$, the prediction $\hat{y}$ for a new data point $x = (x_1, \dots, x_d)$ is

$$\frac{\hat{y} - \bar{y}}{s_y} = \sum_{j=1}^d \frac{x_j - \bar{x}_j}{s_j} \hat{\beta}_j \quad \Leftrightarrow \quad \hat{y} = s_y \sum_{j=1}^d \frac{x_j - \bar{x}_j}{s_j} \hat{\beta}_j + \bar{y}$$

Standardization is unnecessary ...

- for ordinary least squares without regularization
- when the original units are meaningful and one wants interpretable coefficients (e.g. a change in outcome per Swiss franc)

Standardizing the input variables is useful ...

- in regularized regression (truncated SVD, ridge, lasso, elastic net) because penalties depend on the magnitude of the coefficients
- for comparing coefficient sizes: standardization makes coefficients more comparable across different input variables

Standardizing the input and target variables is useful ...

- if one wants standardized regression coefficients
- if one is comparing models across targets with different scales
- since in this case, the intercept is not optimized, so regularization does not penalize the intercept

2.7 Cross-Validation

- Cross-validation refers to a family of *model validation methods* for assessing how a trained model generalizes to unseen data
- One particular cross-validation method is *k-fold cross-validation*
 - it (*randomly*) partitions the training data $(x_i, y_i)_{i=1}^m$ into k subgroups
 - a single subgroup is retained as *validation data* and the data in the remaining $k - 1$ subgroups is used to train the model
 - the process is repeated k times, each time the empirical loss on the validation set is computed
 - the *average validation loss* over the k subgroups is an estimate of the *generalization error*
- It can be used to determine optimal regularization parameters (*such as the threshold in truncated pseudoinversion or the penalty parameter in ridge and LASSO regression*)
(*pick the regularization parameter that leads to the smallest average validation loss*)
- If cross-validation is used to tune *hyperparameters*, the resulting *prediction function* should still be evaluated on an independent *test data set*

2.8 Numerical & Categorical Data

- **Numerical data:** data in $\mathbb{N}$ or $\mathbb{R}$ for which the natural order has a meaning
 - *discrete numerical data:* e.g. number of children, number of car accidents in Switzerland in 2025
 - *continuous numerical data:* e.g. blood pressure, height, BMI (body mass index)
- **Categorical data:** discrete data without a natural order, e.g.
 - married, unmarried, widowed
 - postal code (1000, 3001, 8092, ...) (*is numeric, but there is no natural order*)
- **Unclear cases:** E.g. {low, medium, high} could be treated as categorical or numerical data depending on the context

2.9 One-Hot & Dummy Encoding

- Assume a categorical variable takes values in $\{1, \dots, k\}$, but the natural order has no meaning
- If there are only *two categories*, A and B , they can be encoded as 0 and 1 without problem
- **One-Hot Encoding** If $k \geq 3$, $\{1, \dots, k\}$ can be embedded in $\mathbb{R}^k$ via

$$1 \mapsto \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad 2 \mapsto \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad k \mapsto \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

- **Dummy Encoding** Alternatively, $\{1, \dots, k\}$ can be embedded in $\mathbb{R}^{k-1}$ via

$$1 \mapsto \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \quad 2 \mapsto \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad k \mapsto \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$$

*often, this is enough (e.g. if there is an intercept in linear regression),
it needs less memory than one-hot-encoding*

- **Random embedding** Sometimes, $\{1, \dots, k\}$ is embedded randomly in $\mathbb{R}^q$ for a $q \leq k$

2.10 Feature Engineering

- Sometimes, the performance of a linear regression can be improved significantly by transforming and combining features
- Original features: $x_1, \dots, x_d \in \mathbb{R}$
- Feature engineering $1, x_1, \dots, x_d, e^{\gamma x_1}, x_1 x_2, |x_2 - m|, (x_2 - m)^2$
- **Example: Mortality**
 - y = probability that someone dies in the next year
 - x_1 = gender (female = 0, male = 1)
 - x_2 = age
 - x_3 = BMI
 - Model $y = \beta_0 + \beta_1 x_1 + \beta_2 e^{\gamma x_2} + \beta_3 |x_3 - 24| + \varepsilon$
- Usually, feature engineering needs *expert knowledge*
- Some ML methods (e.g. NN) can be seen as *automatic feature engineering*

2.11 Data Cleaning, Missing Data & Imputation

Data Cleaning

- Remove duplicate or irrelevant observations
- Correct typos, inconsistent capitalization, or improper naming conventions (e.g., standardizing "N/A", "NA" and "Not Applicable" to a single format)
- Identify extreme values that could skew statistics. Outliers might be legitimate edge cases or errors (like an age entered as 250)

Missing Data & Imputation If a row vector $(x_{i1}, \dots, x_{id})$ has a missing component x_{ij} , one can ...

- 1 remove the whole row $\rightsquigarrow$ less data!
- 2 impute the data by ...
 - using the average of $x_{i'j}$ over all $i' = 1, \dots, m$ for which one has data (for numerical data)
 - using the most frequent category taken by $x_{i'j}$, $i' \neq i$ (for categorical data)
 - predicting x_{ij} from existing data $x_{ij'}, j' \neq j$
 - drawing x_{ij} from a learned conditional distribution given existing data $x_{ij'}, j' \neq j$